

HOMEWORK 6
ELEMENTARY LINEAR ALGEBRA
(MATH 2250-03), SPRING, 2018

CHECK: 04/05/2019

SECTION 4.1 - VECTOR SPACES AND SUBSPACES

1. Let V be the set of all polynomials in $\mathbb{P}_n$ such that $p(0) = 0$, i.e.

$$V = \{p \in \mathbb{P}_n : p(0) = 0\}.$$

Show that V is a subspace of $\mathbb{P}_n$.

2. The set of all continuous real-valued functions defined on a closed interval $[a, b]$ in $\mathbb{R}$ is denoted by $C[a, b]$. This set is a subspace of the vector space of all real-valued functions defined on $[a, b]$ (this vector space is infinite-dimensional, but don't let that concern you). Let $V = \{f \in C[a, b] : f(a) = f(b)\}$. Show that V is a subspace of $C[a, b]$.
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SECTION 4.2 - NULL SPACES, COLUMN SPACES, AND LINEAR TRANSFORMATIONS

3. Let $M_{2 \times 2}(\mathbb{R})$ be the vector space of all 2×2 matrices over $\mathbb{R}$, and define $T : M_{2 \times 2}(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ by $T(A) = A + A^T$. Show that T is a linear transformation and give a description of the kernel of T .
4. Define $T : C[0, 1] \rightarrow C[0, 1]$ as follows: For $f \in C[0, 1]$, let $T(f)$ be the antiderivative, F , of f such that $F(0) = 0$. Show that T is a linear transformation and give a description of the kernel of T .
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SECTION 4.3 - LINEARLY INDEPENDENT SETS, BASES

5. Consider the polynomials $p_1(t) = 1 + t^2$ and $p_2(t) = 1 - t^2$. Is $\{p_1, p_2\}$ a linearly independent set in $\mathbb{P}_3$? Why or why not?
6. In the vector space of all real-valued functions, find a basis for the subspace spanned by $\{\sin(t), \sin(2t), \sin(t)\cos(t)\}$. (HINT: Look up the double-angle identities.)
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SECTION 4.4 - COORDINATE SYSTEMS

7. The set $\mathcal{B} = \left\{ \begin{bmatrix} 2 \\ -9 \end{bmatrix}, \begin{bmatrix} 1 \\ 8 \end{bmatrix} \right\}$ is a basis for $\mathbb{R}^2$. Find the change-of-coordinates matrix from the basis $\mathcal{B}$ to the standard basis of $\mathbb{R}^n$.
8. Let $p_1(t) = 1 + t^2$, $p_2(t) = t - 3t^2$, and $p_3(t) = 1 + t - 3t^2$. The set $\mathcal{B} = \{p_1, p_2, p_3\}$ forms a basis for $\mathbb{P}_2$. Find the polynomial $q \in \mathbb{P}_2$ given that $[q]_{\mathcal{B}} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$.